

GALLANT IAS ACADEMY
PRELIMS CUM MAINS BATCH 2027
GEOGRAPHY
CLIMATOLOGY
MODULE 2
EARTH'S ATMOSPHERE AND STRUCTURE

CLIMATOLOGY

Climatology encompasses the study of structure of atmosphere and elements of weather and climates and climatic types and regions.

ATMOSPHERE

The atmosphere is a gaseous envelope surrounding the Earth and is attached to the Earth's surface by the gravitational force. The height of the atmosphere is estimated to range between 16 to 29 thousand kilometres from the sea level. The density of the atmosphere decreases rapidly with altitude - **99% of the total mass of the atmosphere is concentrated in its first 32 kilometres.**

The atmosphere is a mixture of several gases present in the atmosphere and has its own weight. This weight exerted by the atmosphere is responsible for atmospheric pressure. (Technically, air pressure is defined as total weight of a mass of column of air above per unit area.) Generally, the atmospheric pressure decreases with altitude. The distribution of air pressure is also controlled by altitude, temperature, air circulation, rotation of the earth, etc.

COMPOSITION OF ATMOSPHERE

The atmosphere is composed of **gases**, **water vapour**, and **particulate matter (aerosols)**.

Particulate Matter (Aerosols)

Particulate matter includes the suspended solid particles and liquid droplets in the atmosphere such as dust particles, salt particles, pollen, smoke and soot, volcanic ash, etc. In simple terms, it includes all the solid particles and liquid droplets present in the air except gases and water vapour. They can originate from natural sources like dust storms or human activities like industrial emissions. Aerosols play a role in cloud formation, atmospheric chemistry, and climate.

Classification of Atmospheric Gases

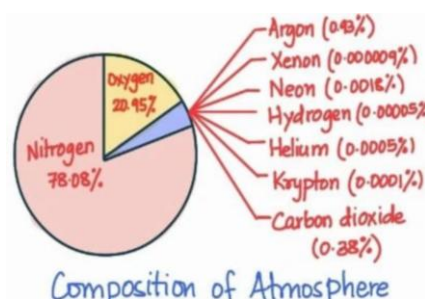
The atmospheric gases are broadly classified into two categories:

<u>Category</u>	<u>Definition</u>	<u>Examples</u>
<u>Constant Gases</u>	Gases whose concentration has remained almost the same in recent history of the Earth. They constitute 99.9% of the atmosphere.	Nitrogen, Oxygen, Argon

<u>Variable Gases</u>	Gases present in small and variable quantities. They have greater control over the environment and climate change.	Carbon dioxide, Methane, Ozone, Water vapour, etc.
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Constant Gases

Nitrogen (78%): It is the most abundant gas in the atmosphere. It is a relatively inert gas produced mainly by volcanic eruptions. It plays a vital role in biological processes and is essential for the formation of proteins and DNA. Through nitrogen fixation, atmospheric nitrogen is transferred into the soils. The roots of the vegetation extract them, while animals get nitrogen from plants by eating them.



Oxygen (21%): It is chemically very active, combining readily with other elements in the process of oxidation. Oxygen enters the atmosphere mainly as a result of photosynthesis by autotrophic green plants of terrestrial ecosystems and phytoplanktons of marine ecosystems. Oxygen plays a very important role in the formation of carbohydrates, fats and proteins. Oxygen is consumed in respiration by animals and in the combustion process.

Argon (0.93%): It is the third most abundant gas in the atmosphere. It is a colourless, odourless, and inert gas.

Variable Gases

Carbon dioxide (0.04%): It is the fourth most abundant gas in the atmosphere. It plays two significant roles. First, it helps in the process of photosynthesis. Second, it is a greenhouse gas that increases the temperature of the atmosphere. Carbon dioxide is mostly transparent to the incoming solar radiation but is opaque for the outgoing terrestrial radiation, thus carbon dioxide increases the greenhouse effect and keeps the surface warm. It is the most significant greenhouse gas. The concentration of carbon dioxide is increasing due to the burning of fossil fuels and deforestation. In the beginning of the industrial revolution (early 19th century) the concentration of atmospheric carbon dioxide was 290 ppm, but it has crossed 420 ppm as of now.

Methane (0.0002%): It is a greenhouse gas which is increasing in the atmosphere due to human activities such as landfills, rice cultivation, natural gas systems, and livestock production. It is twenty times more potent than carbon dioxide as a greenhouse gas.

Ozone: It is both beneficial and harmful to life on Earth. Much of the ozone is found in the stratosphere. Here ozone absorbs the U.V lights from the sun thereby preventing it from reaching the surface. Ozone is also found in the troposphere, where it acts as an eye and respiratory irritant. While ozone in the stratosphere helps shield us from harmful UV radiation, ozone in the troposphere acts as a greenhouse gas, trapping heat and contributing to climate change.

Water Vapour: Even though it is considered in the category of atmospheric gases, it needs separate consideration because it is a very important constituent of the

atmosphere and is responsible for different forms of condensation and precipitation such as clouds, fogs, rainfall, etc. (crucial role in weather and climate). The process of evaporation is responsible for the transformation of water into vapour. It is the most variable of atmospheric gases - it can range from 0.02% in polar regions to 5% in humid tropical regions. It is also a greenhouse gas, absorbing a good portion of the outgoing terrestrial radiation. It can absorb some portion of the incoming solar radiation as well (though very little). The concentration of water vapour also decreases with increasing altitude.

Greenhouse Gases

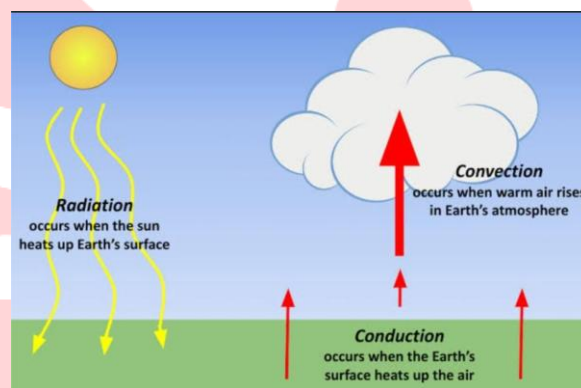
Greenhouse Gases consist of **Carbon dioxide (CO₂)**, **Methane (CH₄)**, **Ozone**, **Nitrous Oxide (N₂O)**, **Chlorofluorocarbons**, and **Water Vapour**. These gases trap the terrestrial radiation and thereby heat the atmosphere, contributing to the greenhouse effect. While vital for maintaining Earth's temperature, increased concentrations can lead to global warming and climate change.

ATMOSPHERIC HEATING

The atmospheric heating occurs mainly through the following processes:

Direct Insolation

The electromagnetic radiations coming from the sun comprise a spectrum of electromagnetic waves. **Insolation** refers to the incoming solar radiation into Earth. It is considered as a shortwave radiation. The atmosphere absorbs some amount of heat energy directly from the incoming shortwave radiations through ozone, water vapour, etc. But this amount is too low to heat the atmosphere significantly.



Conduction

During the daytime, the Earth absorbs heat energy from the sun and gets heated up. The air in contact with the land also gets heated and the process is called **conduction**. The heating through conduction is effective only up to a few meters in the lower atmosphere, thus only the lower atmosphere is heated by this mechanism.

Convection

The transfer of heat energy through the movement of a mass of substance from one place to another is called **convection** (effective mostly in liquids and gases). The air in contact with the Earth rises vertically on heating in the form of air currents and transfers heat energy further into the atmosphere (warm air is lighter and rises). In simple terms, this process of vertical heating is called convection. Horizontal transfer of heat energy (horizontal convection) is called **advection**.

Terrestrial Radiation

The Earth is heated up by the short-wave radiation from the sun. It then radiates back heat energy in the form of **long-wave radiations** (mostly infrared radiations). These radiations are absorbed by the atmospheric gases such as carbon dioxide and water vapour (greenhouse gases) and heat up the atmosphere. Even though the atmosphere is more or less transparent to the incoming shortwave solar radiation,

it absorbs most of the outgoing long wave terrestrial radiations. Thus, **the terrestrial radiation is the most important source of atmospheric heating.** Water vapour absorbs both the incoming shortwave solar radiation and outgoing longwave terrestrial radiation.

Latent Heat of Condensation

During the condensation process, latent heat is released into the atmosphere which further heats up the atmosphere.

STRUCTURE OF ATMOSPHERE

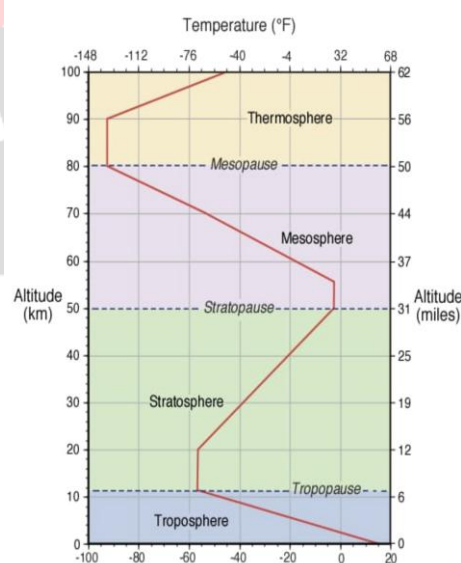
Based on temperature changes, the atmosphere can be depicted as follows.

<u>Layer</u>	<u>Altitude</u>	<u>Temperature Behaviour</u>	<u>Key Feature</u>
<u>Troposphere</u>	0 to 8-18 km	Decreases with altitude (Normal Lapse Rate: 6.5 degrees C/km or about 1 degree C/165m)	All weather; water vapour and clouds; most important for life
<u>Stratosphere</u>	Up to ~50 km	Increases with altitude (Temperature Inversion)	Ozone layer; ideal flying zone
<u>Mesosphere</u>	Up to ~80 km	Decreases with altitude; coldest layer (up to -100 degrees C)	Meteorites burn up here; noctilucent clouds
<u>Thermosphere</u>	80 km and above	Increases with altitude	ISS orbits here; auroras; ionosphere and exosphere

1. Troposphere

The troposphere is the **lowest layer of the atmosphere.** Temperature decreases with altitude in the troposphere as the atmosphere is mostly heated from below. Warm air is lighter (low density); therefore, it rises. As it rises, it gets cooled because of **adiabatic expansion.** During adiabatic expansion, the air expands and the temperature decreases without any heat transfer to the surroundings. The rate at which the temperature decreases with altitude is called the **normal lapse rate** and it is about **6.5 degrees C/km** (or about 1 degree C/165m).

The troposphere is called so as it is the zone of atmosphere in which air keeps turning. The warm air rises, gets cooled and descends back. The word troposphere comes from the Greek word 'tropos' which refers to turning ('mixing' or 'turbulence'). The troposphere extends from the Earth's surface to approximately **8-18 kilometres above sea level**, depending on



the latitude and season. The height of the troposphere is more over the equator because of the higher convection associated with the equatorial regions. Most of what comprises weather remains confined within the troposphere. Almost all water vapour and clouds are found in this layer.

The zone separating the troposphere from stratosphere is known as the **tropopause**. The air temperature at the tropopause is about -80 degrees C over the equator and about -45 degrees C over the poles. Tropopause forms a limit of all upward convection currents.

2. Stratosphere

The stratosphere lies above the troposphere, reaching an altitude of about **50 kilometres**. In the stratosphere lies the **ozone layer**. Much of the ozone exists between 15 to 50 kilometres and the maximum concentration is between 25 to 30 kilometres. Unlike the troposphere, temperature in the stratosphere **increases with altitude** (temperature inversion) due to the presence of the ozone layer, which absorbs ultra-violet radiation and shields life on the earth from intense, harmful forms of energy. Temperature inversion refers to the atmospheric condition in which temperature is increasing with height. At the stratopause, the temperature can become as high as 0 degrees C.

Generally, in the stratosphere there are fewer clouds, but some weather phenomena also occur here. **Contrail clouds** are seen - line-shaped clouds produced by aircraft engine exhaust. The water vapour present in aircraft engine exhaust can condense at low temperatures. They are composed primarily of water in the form of ice crystals. Sometimes cirrus clouds, known as '**mother-of-pearl clouds**' or '**nacreous clouds**', are created in the lower stratosphere (mostly found over polar regions in winter season and therefore also called polar stratospheric clouds).

The Stratosphere is considered as an ideal flying zone for aircrafts. Many commercial jet aircraft fly within the troposphere only, but they will usually move just above it into the lower stratosphere to avoid bad weather and turbulence for a smooth flight. The lower part of the stratosphere, just above the tropopause, acts as an ideal flying zone for three reasons: the relative absence of clouds, the relative absence of strong convective currents, and isothermal conditions.

3. Mesosphere

Above the stratosphere is the mesosphere, extending up to approximately **80 kilometres**. Mesosphere is a transition zone between the stratosphere and the thermosphere and is heated by the stratosphere. In this layer, once again, temperature starts decreasing with the increase in altitude and reaches up to **-100 degrees C at the height of 80 km**, making it the **coldest layer of the atmosphere**. Mesopause has the lowest temperature in the entire atmosphere. Meteorites burn up in this layer on entering from space. **Noctilucent clouds** are formed when moisture condenses on meteoric dust particles (mostly found over polar regions in summer).

4. Thermosphere

The thermosphere begins around **80 kilometres**. Due to the selective absorption of short-wave radiation, temperature here starts increasing with height. This is the layer where the **International Space Station** and satellites mostly orbit. The **auroras** occur in this layer due to interactions between solar wind (charged particles such as electrons released from the Sun) and atmospheric gas particles. Auroras

are glowing lights seen in the polar sky during midnight - commonly known as the northern lights (**aurora borealis**) or southern lights (**aurora australis**). The thermosphere is divided into the **ionosphere** and **exosphere**.

a) Ionosphere

Discovered by **Kennelly and Heaviside**. In the lower part of the thermosphere, the ionosphere is found. It contains electrically charged particles known as ions, and hence is known as the ionosphere. Ionization of molecules and atoms occurs in the ionosphere by UV rays, X-rays, and Gamma rays (for example, X Rays are absorbed by N₂ and O₂ which causes their ionization). This layer shields the Earth from harmful high energy radiations. Radio waves transmitted from the earth are reflected to the earth by this layer, therefore it helps in broadcasting signals.

There are several ionic layers within this layer:

<u>Layer</u>	<u>Behaviour</u>	<u>Notes</u>
<u>D Layer</u>	Reflects low frequency radio waves; absorbs medium and high frequency radio waves.	Disappears with sunset.
<u>E Layer</u>	(Kennelly-Heaviside layer) Reflects medium and high frequency radio waves.	Also disappears with sunset.
<u>F Layer</u>	(Appleton layer) Consists of two sub-layers F1 and F2. Reflects medium and high frequency radio waves back to earth.	F1 disappears at night. F2 remains day and night - responsible for most long-distance radio communication.

b) Exosphere

The uppermost layer of the atmosphere is known as the **exosphere** (also called 'outer atmosphere'). It is an extremely rarefied layer (very thin air), with very few gas particles being present, and it gradually merges with outer space.

CHEMICAL COMPOSITION OF THE ATMOSPHERE

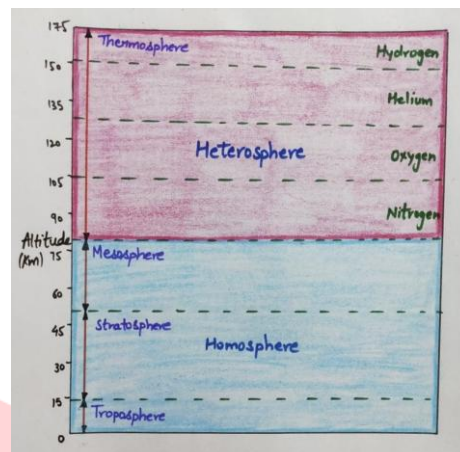
Based on chemical composition, the atmosphere is divided into **homosphere** and **heterosphere**.

<u>Feature</u>	<u>Homosphere</u>	<u>Heterosphere</u>
<u>Extent</u>	Surface to 80 km	80 km and above
<u>Gas Mixing</u>	Gases are uniformly mixed; proportions of different gases are uniform at different levels.	Gases are no longer evenly mixed; begin to separate into sub-layers of concentration.
<u>Main Constituents</u>	Oxygen (21%) and Nitrogen (78%), along with Argon, CO ₂ , Neon,	Molecular nitrogen (80-200 km), Atomic oxygen (200-1100 km), Helium (up to

	Helium, Krypton, Xenon, Hydrogen, etc.	3500 km), Atomic hydrogen (topmost layer).
Thermal Layers	Divided into: Troposphere, Stratosphere, Mesosphere.	Properties vary significantly by sub-layer.

1. Homosphere

The Homosphere represents the lower portion of the atmosphere and extends up to the height of **80 km from sea level**. This zone is called homosphere because of the homogeneity of the proportion of various gases - in other words, the gases are uniformly mixed. Man is increasingly disturbing the natural proportions of gases through ever-increasing economic activities and modern technologies. For example, the proportion of carbon dioxide is rapidly increasing due to burning of fossil fuels (coal, petroleum and natural gas) and deforestation. On the other hand, the proportion of ozone gas was rapidly decreasing due to ever increasing production and consumption of CFCs (chloro-fluorocarbons) and halogenated gases. Based on thermal conditions, the homosphere is divided into three layers: (i) troposphere, (ii) stratosphere, and (iii) mesosphere.



2. Heterosphere

The heterosphere extends from **80 km and above**. Here the gases are no longer evenly mixed and begin to separate into sub-layers of concentration. Different layers of this sphere vary in their chemical and physical properties. In this layer, the heavier gases are pulled closer to the Earth's surface by the Earth's gravity. There are four distinct layers of gases in this sphere:

<u>Layer</u>	<u>Extent</u>
<u>Molecular Nitrogen Layer</u>	80 to 200 km
<u>Atomic Oxygen Layer</u>	200 to 1100 km
<u>Helium Layer</u>	Up to 3500 km
<u>Atomic Hydrogen Layer</u>	Topmost layer - extends up to the outermost limit of the atmosphere

PRELIMS PREVIOUS YEAR QUESTIONS

1. Consider the following statements: [2025]
 - I. Without the atmosphere, temperature would be well below freezing everywhere on the Earth's surface.
 - II. Heat absorbed and trapped by the atmosphere maintains our planet's average temperature.
 - III. Atmosphere's gases, like carbon dioxide, are particularly good at absorbing and trapping radiation.Which of the statements given above are correct?
 - (a) I and III only
 - (b) I and II only
 - (c) I, II and III
 - (d) II and III only
2. Consider the following statements: [2024]

Statement-I: Thickness of the troposphere at the equator is much greater as compared to poles.

Statement-II: At the equator, heat is transported to great heights by strong convectional currents.

Which one of the following is correct in respect of the above statements?
 - (a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I
 - (b) Both Statement-I and Statement-II are correct, but Statement-II does not explain Statement-I
 - (c) Statement-I is correct, but Statement-II is incorrect
 - (d) Statement-I is incorrect, but Statement-II is correct
3. With reference to "water vapour", which of the following statements is/are correct? [2024]
 1. It is a gas, the amount of which decreases with altitude.
 2. Its percentage is maximum at the poles.Select the answer using the code given below:
 - (a) 1 only
 - (b) 2 only
 - (c) Both 1 and 2
 - (d) Neither 1 nor 2
4. Normally, the temperature decreases with the increase in height from the Earth's surface, because: [2012]
 1. The atmosphere can be heated upwards only from the Earth's surface
 2. There is more moisture in the upper atmosphere
 3. The air is less dense in the upper atmosphereSelect the correct answer using the codes given below:
 - (a) 1 only
 - (b) 2 & 3 only
 - (c) 1 & 3 only
 - (d) 1, 2 & 3
5. A layer in the earth's atmosphere called the ionosphere facilitates radio communication. Why? [2011]
 1. The presence of ozone causes the reflection of radio waves to earth.
 2. Radio waves have a very long wavelength.Which of the statements given above is/are correct?
 - (a) 1 only
 - (b) 2 only

- (c) Both 1 and 2
(d) Neither 1 nor 2
6. The correct sequence of different layers of the atmosphere from the surface of the Earth upwards is: [1998]
(a) Troposphere, Stratosphere, Ionosphere, Mesosphere
(b) Stratosphere, Troposphere, Ionosphere, Mesosphere
(c) Troposphere, Stratosphere, Mesosphere, Ionosphere
(d) Stratosphere, Troposphere, Mesosphere, Ionosphere
7. Which one of the following layers of the atmosphere is responsible for the deflection of radio waves? [1995]
(a) Troposphere
(b) Stratosphere
(c) Mesosphere
(d) Ionosphere
8. The jet aircrafts fly very easily and smoothly in the lower stratosphere. What could be the appropriate explanation? [1995]
1. There are no clouds or water vapour in the lower stratosphere.
2. There are no vertical winds in the lower stratosphere.
Which of the statements given above is/are correct in this context?
(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

MAINS PREVIOUS YEAR QUESTIONS

"The troposphere is a very significant atmospheric layer that determines weather processes. How?" (250 words, 15 marks) (2022)

NCERT References

Class 11th: Fundamentals of Physical Geography	Chapter 7 - Composition and Structure of Atmosphere Chapter 8 - Solar Radiation, Heat Balance and Temperature
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